

Title:

Console-Based Mini Project – Student Management System

1. Objective

Build a simple Java console application using loops and conditions.

2. Features

- Add student
- View students
- Delete student

3. Code

```
import java.util.*;
```

```
class Student {
```

```
    int id;
```

```
    String name;
```

```
    Student(int id, String name) {
```

```
        this.id = id;
```

```
        this.name = name;
```

```
    }
```

```
}
```

```
public class StudentManagement {
```

```
    public static void main(String[] args) {
```

```
        ArrayList<Student> list = new ArrayList<>();
```

```
        Scanner sc = new Scanner(System.in);
```

```
        while (true) {
```

```
System.out.println("\n--- Student Management System ---");
```

```
System.out.println("1. Add Student");
```

```
System.out.println("2. View Students");
```

```
System.out.println("3. Delete Student");
```

```
System.out.println("4. Exit");
```

```
System.out.print("Enter choice: ");
```

```
int choice = sc.nextInt();
```

```
if (choice == 1) {
```

```
    System.out.print("Enter ID: ");
```

```
    int id = sc.nextInt();
```

```
    sc.nextLine();
```

```
    System.out.print("Enter Name: ");
```

```
    String name = sc.nextLine();
```

```
    list.add(new Student(id, name));
```

```
    System.out.println("Student added!");
```

```
} else if (choice == 2) {
```

```
    if (list.isEmpty()) {
```

```
        System.out.println("No students found.");
```

```
    } else {
```

```
        for (Student s : list) {
```

```
            System.out.println("ID: " + s.id + ", Name: " + s.name);
```

```
        }
```

```
    }
```

```
} else if (choice == 3) {
```

```
    System.out.print("Enter ID to delete: ");
```

```
    int id = sc.nextInt();
```

```
boolean found = false;
for (Student s : list) {
    if (s.id == id) {
        list.remove(s);
        found = true;
        System.out.println("Student deleted!");
        break;
    }
}

if (!found) {
    System.out.println("Student not found.");
}

} else if (choice == 4) {
    System.out.println("Exiting...");
    break;

} else {
    System.out.println("Invalid choice!");
}
}
}
```

4. Output

```
C:\Windows\System32\cmd.e  X  +  v

C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>java StudentManagement

--- Student Management System ---
1. Add Student
2. View Students
3. Delete Student
4. Exit
Enter choice: 1
Enter ID: 25042
Enter Name: sushanth
Student added!

--- Student Management System ---
1. Add Student
2. View Students
3. Delete Student
4. Exit
Enter choice: 2
ID: 25042, Name: sushanth

--- Student Management System ---
1. Add Student
2. View Students
3. Delete Student
4. Exit
Enter choice: 4
Exiting...

C:\Users\susha\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\New folder>
```

Conclusion:

This project demonstrates real-world use of Java basics.